EchoPin: Implementation Phase

Project: EchoPin, privacy-first group presence over Bluetooth and Wi-Fi, with an optional central server
Version documented: 0.5.0 (app), 1.1.0 (server) **Repository:** https://github.com/BEEKONTek/EchoPin
Author: Botlhale Pono Maruma **Date:** 12 November 2026

This document covers the implementation phase: how the system was coded, how it was tested, the system test cases with their results, and how it is installed. It refers to the live repository for every claim, so each statement can be checked.

5.1 Introduction

EchoPin keeps a small group together without tracking anyone. Phones in a **session** see each other as distances on a pinwheel, raise alerts when someone crosses an agreed boundary, check in on each other, and send an SOS that every member receives, all phone to phone over Bluetooth Low Energy and Wi-Fi. No coordinate is ever measured, stored or sent; distance comes from signal strength on the receiving phone and is deliberately jittered before it reaches the screen.

The implementation phase delivered:

1. **The app**, for Android, iOS, Linux, macOS, Windows and the web (demo only), built with Flutter 3.44 and Dart 3.12.
2. **A shared protocol package** (`packages/echopin_protocol`) holding the sealed-box cryptography, device proofs, wire enumerations and the input limits used by both app and server.
3. **The optional server** (`server/`), Dart on PostgreSQL, adding verified accounts, device-bound sign-in, sealed emergency records that only an authorised responder can open, reports and diagnostics from every device, a staff console, an append-only audit chain and encrypted backups. A live instance runs on free hosting (Render, Supabase, Brevo).
4. **Tooling**: continuous integration for lint, tests and builds on every push, a one-command server test suite, a one-command system test, an installer that works by QR code, and a web demo published to GitHub Pages.

The guiding constraints throughout were: the app must stay fully usable with no server; no location data may exist anywhere in the system; every failure must be explained to the person in plain words; and the server must never trust the app.

Related documents: `README.md` (overview), `docs/ARCHITECTURE.md` (diagrams), `docs/CODE_MAP.md` (file-by-file guide), `docs/PROTOCOL.md` (wire format), `docs/THREAT_MODEL.md` and `docs/SECURITY_REVIEW.md` (security), `docs/SERVER.md` and `docs/DEPLOY_HOSTED.md` (operations).

5.2 Coding

5.2.1 Languages, frameworks and tools

Part	Technology	Why
App	Flutter 3.44 / Dart 3.12	one code base for six platforms; all cryptography in Dart so it is identical everywhere and testable on a desktop
Native plugins	Kotlin (Android), Swift (iOS)	Bluetooth LE advertising, scanning and GATT; they move opaque bytes only
Shared package	Dart (`package:cryptography`)	X25519 + HKDF + XChaCha20-Poly1305 sealed boxes, Ed25519 device proofs
Server	Dart (`shelf`, `postgres`)	same language as the app, one team, shared models
Database	PostgreSQL 16–18	row-level security, triggers for an append-only audit log
Email	Brevo HTTP API (SMTP and Resend supported)	free tier; Render's free plan blocks SMTP
Hosting	Render (Docker), Supabase (PostgreSQL + Storage)	free, HTTPS, daily encrypted backups
CI	GitHub Actions	lint, tests, builds for every platform, Pages deployment

5.2.2 Code structure

```
lib/                     the app
  core/                  EchoError: every failure as a typed, human sentence
  security/              invite code → keys (HKDF), Ed25519 identity, envelopes,
                         SecureMessenger (decrypt, verify, pin, replay, rate limit)
  session/               SessionEngine (membership, pings, SOS, receipts, retries)
  transport/             Wi-Fi (Bonjour + TCP), in-memory loopback
  services/              Bluetooth transport + bridge, MeshService, noise, alerts,
                         notifications, connectivity, history, settings, PIN auth
  providers/             ChangeNotifiers: Session, Alert, Settings, Admin,
                         Account, AccountLock, Outbox
  account/               server client, AccountProvider, Outbox, SessionRecorder
  screens/ widgets/ ui/  home, session, pinwheel, trays, admin, account, console
android/ ios/            native BLE plugins
packages/echopin_protocol sealed boxes, device proofs, enums, ProtocolLimits
server/
  lib/src/api/           http.dart (parsing, validation, error mapping), routes.dart
  lib/src/auth/          AuthService, passwords (Argon2id), TOTP, breach check
  lib/src/               AdminService, TelemetryService, RetentionService
  lib/src/security/      keys, audit chain, custodian keys, rate limits
  lib/src/backup/        encrypted backups, storage backends
  lib/src/db/            pool and migrations (7 so far)
  bin/                   server.dart, echopin_admin.dart (operator CLI)
  tool/                  test.sh, live_check.sh + live_check.dart (system test)
scripts/                 share_apk.sh (install by QR code)
```

5.2.3 Coding standards

- **Analyzer and formatter are law.** `flutter analyze` must report no issues across all three packages and `dart format` must produce no change; CI fails otherwise (`.github/workflows/lint.yml`).
- **Lints**: `flutter_lints` plus the repository's `analysis_options.yaml` (typed locals, braces on every control statement, no print statements in library code).
- **Errors carry a sentence.** In the app every failure is an `EchoError` subclass with a human message and, where possible, a one-tap fix. On the server every expected failure is an `ApiException(status, code, message)`; clients switch on the code and show the message. Stack traces and database wording never reach a client.
- **Comments explain why.** Each security-relevant block starts with a short note of the threat it addresses (`SECURITY:` tags in auth code).
- **Shared constants, not copies.** Text limits live once in `ProtocolLimits` and are read by the app's fields and the server's checks.
- **Version control.** Conventional commit messages (`feat:`, `fix:`, `docs:`); `main` is the release branch and `update` the working branch; every push runs the full CI.

5.2.4 Key modules

Module	Responsibility	Notable decisions
`security/session_credentials.dart`	invite code ↔ session id and key	16 base-32 characters (~78 bits) plus a 2-character checksum that works offline on any device; the checksum catches typos before any crypto runs
`security/secure_messenger.dart`	envelope decrypt/verify, sender pinning, replay window, rate limits	sender ids are derived from public keys and pinned on first sight, so an impersonator is detected at the second message
`session/session_engine.dart`	hello / ping / receipt / reply / bye, retries, SOS lifecycle	pings retry until acknowledged; SOS state is replayed to late joiners and resolved everywhere on *I'm safe*
`services/location_engine.dart`	privacy noise	jitter is fixed per peer per minute (FNV-1a seeded), so repeated samples cannot be averaged away
`widgets/pinwheel_painter.dart`	the radar	100 m scale; peers without a measured bearing are spread evenly and the caption says *direction unknown*, so no direction is ever invented
`account/account_provider.dart`	sign-up, sign-in, tokens, device key, two-step, custodian keys	offline-first: a saved sign-in survives restarts; local pre-checks on email, name, password and codes before any request
`providers/account_lock_provider.dart`	PIN in front of the signed-in account	PIN per sign-in, 5-minute idle lock on the wall clock, third lockout signs out

Module	Responsibility	Notable decisions
`account/outbox.dart`, `session_recorder.dart`	sealed emergency records	sealed to the custodian key, then the escrow key; checkpoint uploaded within seconds of an SOS; queued offline
`server/lib/src/auth/auth_service.dart`	accounts	Argon2id (19 MiB, t=2), uniform response time, per-IP/email/device limits in the database, known-device bypass so a stranger cannot lock the owner out, 8-digit codes with a daily cap
`server/lib/src/admin_service.dart`	console and break-glass	roles re-read on every request; unseal writes the grant and audit row before decrypting and emails every admin
`server/lib/src/security/audit.dart`	audit chain	HMAC-SHA256 chain keyed from a secret outside the database; UPDATE/DELETE/TRUNCATE refused by triggers; chain tip anchored with every backup
`server/lib/src/backup/`	backups	`pg_dump` → AES-256-GCM 1 MiB frames → private Supabase bucket in 40 MB parts; verify and restore from the CLI

5.2.5 Input validation and error handling

Every piece of text a person or a device can send is bounded twice: in the app, so the limit is visible while typing, and on the server, which never trusts a client. The limits are one set of constants (`ProtocolLimits`) used on both sides.

Input	Where it is entered	App check	Server check
Your name (session)	Home	max 24 characters, control/bidi characters stripped from network copies	n/a (phone to phone); peers strip and cap received names
Session name	Home	max 40	n/a
Invite code	Home, QR, link	checksum, alphabet (no `0`, `1`), length; precise message per mistake	n/a
Account name	Sign-up	max 40, not empty	`cleanDisplayName`: control characters removed, max 40
Email	Sign-in, sign-up, reset	max 254, shape `x@y.z`	`normalizeEmail`: RFC-shaped regex, lower-cased, max 254
Password	Sign-in, sign-up, change, reset	max 256; sign-up: at least 10, not your email, not a common pattern	same policy, plus a breached-password lookup (HIBP k-anonymity, fails open)
Email code	Verify, reset	exactly 8 digits	constant-time compare, attempt cap, expiry, one generic error

Input	Where it is entered	App check	Server check
Authenticator code	Sign-in, two-step setup	exactly 6 digits	RFC 6238 window, replay refused
Account / admin PIN	Lock screens	4–12 digits	local only; PBKDF2, lockout, lockout count
Report subject / body	Report a problem	max 120 / 4000	same (`ProtocolLimits`)
Unseal reason / case reference	Console	max 1000 / 100	same
Report note / resolution	Console	max 4000 / 2000	same
Search text	Console	max 200	400 if longer
Custodian key	Console	prefix `ECHOPIN-CUSTODIAN-1:`, base64url seed of 32 bytes	n/a (never sent to the server)
Ids in URLs	app → server	n/a	`uuidParam` / `idParam`: anything that cannot be an id answers 404 before any query
Paging (`limit`, `offset`, `before`)	app → server	n/a	clamped to 1–500 / ≥ 0
Request body	app → server	n/a	≤ 64 KB, must be a JSON object; typed field accessors give precise 400s
Sealed uploads	app → server	sealed on the phone	≤ 512 KB, key id must match a known escrow key, custodian seal required when configured, idempotent by id
Server settings	operator	n/a	refused at start-up with a message naming the variable (bad key bundle, dashboard URL pasted as API URL, CORS `*` outside development…)

Error handling follows one rule on each side:

- **App.** Network and server failures become `OfflineError` or `AccountError(code, message)`; screens show `errorText(e)` and keep the person's input. Background work (`unawaited`) always catches and either audits or logs; the twelve deliberately silent catches in the code base are all shutdown or best-effort paths (closing radios, deleting a key that may not exist) and are commented as such.
- **Server.** One middleware (`errorsAndLogging`) converts every exception: `ApiException` → its status and code; a PostgreSQL *data* error (SQLSTATE class 22) → 400 without the database's wording; a uniqueness clash → 409; anything else → 500 with a generic message and a server-side log line (never the request body). Failures in mail and backups are recorded in the audit log and the backups table, so the console shows them without access to host logs.

5.3 Testing

5.3.1 Strategy

Testing runs at six levels, from pure functions to the live deployment. Everything below unit level uses real components: real cryptography, a real in-memory radio, a real PostgreSQL, a real HTTP server, the real Linux build of the app.

Level	What runs	Where	Count
Unit	codes, key derivation, envelopes, PBKDF2 vectors, noise, models, settings migration	`test/security`, `test/models`, `test/services`	103
Component	SessionEngine with lossy and partitioned links, providers over a shared radio, account provider against a fake server, outbox and recorder	`test/session`, `test/providers`, `test/account`	89
Widget (end-to-end UI)	home, demo with answers, hold-to-send SOS, admin and account locks, sign-in and two-step, report filing, console, input limits	`test/widgets`	32
Protocol package	sealed boxes, device proofs	`packages/echopin_protocol/test`	6
Server	every endpoint against a throwaway PostgreSQL: registration, codes, sign-in, tokens, two-step, roles, uploads, break-glass, audit chain, backups and restore, rate limits, hardening regressions, input validation	`server/test`	75
Integration	real Bonjour discovery and TCP between two engines; the real app driven through sign-up, two-step, console and break-glass against a real server	`integration_test/`	3
System	31 checks over HTTP against a freshly started server	`server/tool/live_check.sh`	31
Live deployment	health, database, email delivery, backups, data-API lock-down on the hosted instance	manual, recorded in §5.4	9

Totals at the time of writing: **239 app tests pass (1 skipped without a server), 75 server tests pass, 6 protocol tests pass, 31 of 31 system checks pass.**

5.3.2 Tools

- `flutter test` with `--coverage` in CI; widget tests use `FakeAsync` and a fake HTTP server (`test/helpers/fake_echopin_server.dart`) that implements the API faithfully, including device proofs, so sign-in flows are exercised end to end without a network.
- `server/tool/test.sh` starts a throwaway PostgreSQL cluster (`initdb` / `pg_ctl`), runs all server tests and tears it down; CI does the same with a PostgreSQL 16 service.
- `server/tool/live_check.sh` starts a real server with fresh keys and a custodian key, creates an admin, and runs `live_check.dart`, which acts as several phones and a console over plain HTTP.

- `integration_test/live_server_test.dart` drives the real Linux build of the app under Xvfb against that server, with screenshots at each step.
- GitHub Actions runs lint, all test levels except the system and live ones, and release builds for every platform on each push, publishing failing test names as annotations.

5.3.3 How a bug found in testing was handled

Each defect found during testing was fixed with a regression test in the same commit. Examples from this phase, all reproduced before fixing:

Found by	Defect	Fix and test
system test	a member-proof tie (one proven, one not) proved both	strict majority of distinct accounts; tie → nobody proven; `api_test` "hardening"
system test	SOS counted once per upload, inflating the chart	`max()` per session, `count(distinct)` for the chart
GUI integration test	two-step dialog crashed inside a layout builder	fixed-size QR box
live deployment	server image had no CA certificates: mail and backups failed	`ca-certificates` in the image
live deployment	Supabase's public API could read and write every table	RLS enabled and grants revoked on every start; regression test creates the roles and proves access is gone
live deployment	`MAIL_FROM` as `Name <address>` refused by Brevo	parsed; provider's reason recorded in the audit log
live deployment	an interrupted backup blocked all later backups	cleanup no longer depends on the store being reachable; store errors recorded on the backup row
live deployment	dashboard URL pasted as `SUPABASE_URL`	normalised; refused with guidance when not a URL
CI	restore tests failed on the runner: `pg_restore` newer than the server	client pinned to the service's major
CI	one timing-sensitive widget test failed only on slow runners	tests wait for the fake server to be idle instead of a fixed delay
CI	the hosted-backup test built its own config and ignored the pinned client, so a PostgreSQL 18 dump went to a 16 server	all test configs share `pgDumpBinary` / `pgRestoreBinary`; reproduced locally against a PostgreSQL 16 container
CI	one check-in answer out of four was lost on fast runners: a group joining in the same instant sends each newcomer a direct hello, every phone overhears them, and the per-sender token bucket was charged for frames not addressed to it	the messenger charges the bucket only for frames addressed to the receiver; regression test in `secure_messenger_test`
review	malformed ids in URLs reached the database and produced 500s	`uuidParam` / `idParam`; `api_test` "input validation"

5.4 System testing

5.4.1 Test environment

- **System under test**: server 1.1.0 built from `main`, PostgreSQL 18 locally (16 in CI, 17 hosted), app 0.5.0 Linux release build.
- **Local**: `server/tool/live_check.sh` (fresh database and keys each run) and the GUI integration test under Xvfb.
- **Hosted**: https://echopin-server-z3tg.onrender.com with Supabase (eu-west-1) and Brevo, checked from the public internet.
- **Phones**: Android phones with Bluetooth LE, app 0.5.0 release APK installed by QR code, tested outdoors and indoors on 13 November 2026.

5.4.2 Test cases and results

A. End-to-end over HTTP (`live_check.sh`, 31 checks, automated)

ID	Area	Case	Expected	Result
TC-01	Registration	password found in a public breach	refused with `breached_password`	Pass
TC-02	Registration	valid sign-up	202 `verification_sent`	Pass
TC-03	Registration	timing of sign-up	≥ 600 ms (uniform)	Pass
TC-04	Enumeration	sign-up with an address already taken	identical 202 and timing	Pass
TC-05	Email codes	code in the email	8 digits	Pass
TC-06	Pre-hijack	right code, different password	`password_mismatch`	Pass
TC-07	Email codes	wrong code	one generic error, no attempt count	Pass
TC-08	Enumeration	wrong code for an unknown address	identical answer	Pass
TC-09	Verification	fresh code + sign-up password	account verified	Pass
TC-10	Sign-in	device-proof sign-in	200 with tokens	Pass
TC-11	Lockout	5 wrong passwords from a new device	423 locked	Pass
TC-12	Lockout	owner's known device during the lock	still signs in	Pass
TC-13	Enumeration	5 wrong passwords for an address with no account	locks identically	Pass

ID	Area	Case	Expected	Result
TC-14	Staff two-step	admin without two-step opens the console	`two_step_required`	Pass
TC-15	Two-step	enable with a wrong password	refused	Pass
TC-16	Two-step	enable with password + code	enabled	Pass
TC-17	Staff two-step	console after enabling	answers	Pass
TC-18	Sealed records	double-sealed upload	accepted	Pass
TC-19	Registration	a second person registers	202	Pass
TC-20	Sign-in	second person signs in	200	Pass
TC-21	Sealed records	single-sealed upload when a custodian is required	`custodian_required`	Pass
TC-22	Membership proof	upload from a non-member	session flagged `unproven`	Pass
TC-23	Membership proof	one member, one non-member	tie: nobody proven, both flagged	Pass
TC-24	Roles	responder views the session	names, no email addresses	Pass
TC-25	Break-glass	responder opens a record with a reason	only the inner seal returned	Pass
TC-26	Custodian	custodian key on the device	record readable	Pass
TC-27	Alerting	after the opening	every admin emailed	Pass
TC-28	Audit	chain verification	keyed HMAC chain and backup anchor verify	Pass
TC-29	Audit	session views and unseals	present in the audit log	Pass

(Checks 15–16 run twice in the script, for two accounts, giving 31 results.)

B. The real app against a real server (`integration_test/live_server_test.dart`, automated under Xvfb)

ID	Case	Expected	Result
TC-30	Sign up in the app, confirm the 8-digit code from the email	Account PIN screen, then the Account screen	Pass

ID	Case	Expected	Result
TC-31	Open the console as an admin without two-step	refused with instructions	Pass
TC-32	Turn on two-step from the real dialog	"Two-step sign-in is on"; console opens	Pass
TC-33	Import the custodian key	"Imported key"	Pass
TC-34	Open a record with a reason	record readable, "Recorded by" shown	Pass

C. Widget-level flows (`flutter test`, automated, selected)

ID	Case	Expected	Result
TC-35	Demo session: check-ins answered, hold-to-send SOS, "I'm safe"	trays and halos update; SOS resolves	Pass
TC-36	Invite code with a typo	precise inline message	Pass
TC-37	Admin PIN: wrong PINs, lockout countdown, auto-lock	locked, countdown, relocks	Pass
TC-38	Account PIN: lock, wrong PIN, unlock, forgot PIN signs out, change PIN, auto-lock keeps a draft	all as specified	Pass
TC-39	Urgent safety report	sent as `critical` with diagnostics	Pass
TC-40	Radar with Bluetooth-only peers	spaced evenly, no "N", caption "direction unknown"	Pass
TC-41	Field limits and local checks	capped at `ProtocolLimits`; bad email or empty password refused with no request sent	Pass

D. Live hosted deployment (manual, 9–10 November 2026)

ID	Case	Expected	Result
TC-42	`GET /v1/health` from the internet	`ok: true, database: up`	Pass
TC-43	Unauthenticated and forged-token requests to `/v1/me`, `/v1/admin/*`	401	Pass
TC-44	CORS preflight from an unknown origin	no `Access-Control-Allow-Origin`	Pass
TC-45	Plain HTTP	301 to HTTPS	Pass
TC-46	Sign-up email through Brevo	code arrives in the inbox	Pass (after the `MAIL_FROM` and certificate fixes)
TC-47	Daily backup to Supabase Storage	encrypted file in a private bucket	Pass (after the certificate, slot and URL fixes)

ID	Case	Expected	Result
TC-48	Supabase data API with the public key	no table readable or writable; RLS on all 14 tables	Pass (after the lock-down fix)
TC-49	Database migrations on deploy	`schema_migrations` at the latest version	Pass
TC-50	Supabase security advisor	no critical findings	Pass

F. Bluetooth LE between Android phones (manual, 13 November 2026)

ID	Case	Expected	Result
TC-51	Two phones, one creates a session, the other joins by QR	both appear in each other's tray within seconds	Pass
TC-52	Join by typed code, including one with a typo	typo refused with a precise message; correct code joins	Pass
TC-53	Walk apart and back	distance on the pinwheel rises and falls; peer card updates live	Pass
TC-54	Distance against a known spacing	figure within a few metres	Pass with limitation: signal strength moves with bodies, walls and how the phone is held, so the figure is close, not exact; the app labels it "about" and jitters it by design
TC-55	Cross the boundary circle	banner, haptic and notification on the other phone	Pass
TC-56	Check in on one person, then everyone	delivered receipts; answers appear in the tray	Pass
TC-57	Hold to send SOS; "I'm safe"	SOS reaches every phone with a halo and notification; resolves everywhere	Pass
TC-58	Compare safety codes and mark verified	codes match; badge shown	Pass
TC-59	Leave and rejoin from history	rejoin without retyping the code	Pass
TC-60	Bluetooth off on one phone	status tray with a one-tap "Turn on"	Pass

E. Continuous integration (every push)

Workflow	Jobs	Result on `main`
Lint	analyze all packages, format check	Pass
Test	app tests with coverage, Linux integration, server + protocol on PostgreSQL 16	Pass (Linux integration best-effort on the runner's mDNS)

Workflow	Jobs	Result on `main`
Build	Android, web, Linux, iOS (no codesign), macOS	Pass; **Windows fails** and is non-blocking
Web demo	build and publish to GitHub Pages	Pass

5.4.3 Evaluation of the results

- **Functional coverage.** Every user-facing flow has at least one automated test at widget level, and every server endpoint is exercised by the server suite; the 31 system checks cover the whole account, record and break-glass lifecycle as real clients would drive it.
- **Security properties hold under test**: no account enumeration through answers or timing (TC-04, TC-08, TC-13), no lock-out of the legitimate owner (TC-12), two-step cannot be enabled with a stolen token (TC-15), the server alone cannot read a record (TC-25/26), every opening is audited and alerted (TC-27/29), and the audit chain verifies against an anchor outside the database (TC-28).
- **Defects found were real and are fixed**: the testing phase found four deployment defects that would have stopped real use (certificates, mail sender, backup slot, exposed data API) and three logic defects (tie proof, SOS count, dialog crash). All have regression tests or are re-checked by the system test.
- **Hardware.** Bluetooth LE between Android phones works as designed for every flow (part F). The one limitation is distance accuracy: a signal-strength estimate cannot be exact, which the design already assumes (the radar says "about", jitters the figure, and a boundary alert is a prompt to look up rather than a claim about metres). Direction is not measured at all and the radar says so.
- **Known gaps**, stated rather than hidden: iOS and macOS builds are compiled in CI but not run on devices; the Windows build fails in CI and no Windows host was available to diagnose it; the Linux mDNS integration test is reliable locally but not on GitHub's runners.

5.5 Installation

5.5.1 Requirements

Target	Needs
Android phone	Android 7.0 (API 24) or newer; Bluetooth LE; about 80 MB free
iPhone	iOS 13 or newer; built and signed on a Mac (see `docs/SETUP.md`)
Linux desktop	GTK 3, `libsecret`, Avahi (Wi-Fi sessions)
Web browser	any modern browser; demo mode only
Server (optional)	Docker, or Dart 3.12 + PostgreSQL 16–18; or the free hosted recipe

5.5.2 Install the app on Android

By QR code (recommended for a group):

```
flutter build apk --release \
  --dart-define=ECHOPIN_SERVER_URL=https://your-server \
```

```
  --dart-define=ECHOPIN_CUSTODIAN_KEY=<custodian public key>
scripts/share_apk.sh
```

The script serves the APK on your Wi-Fi and prints a QR code. On each phone: scan it with the camera, tap **Download**, open the file, allow installs from the browser if asked, tap **Install**. Press Ctrl+C when done; the script closes the firewall port it opened.

By cable:

```
adb install -r build/app/outputs/flutter-apk/app-release.apk
```

Omit the two `--dart-define`s for a fully local build with no accounts.

5.5.3 Install on Linux

```
 flutter build linux --release
tar -C build/linux/x64/release -czf EchoPin-linux-x64.tar.gz --transform
's,^bundle,EchoPin,' bundle
```

On the target machine: extract and run `EchoPin/echo_pin`. Keep the `lib/` and `data/` folders beside the executable.

5.5.4 Web demo

Published automatically from `main` to https://beekontek.github.io/EchoPin/ by `.github/workflows/pages.yml`. To host elsewhere: `flutter build web --release --base-href /your/path/` and serve `build/web/`.

5.5.5 Install the server

Free hosted (about an afternoon), step by step in `docs/DEPLOY_HOSTED.md`:

1. Create a Supabase project (database + storage), a Brevo account with a verified sender, and a Render account.
2. From `server/`: `dart run bin/echopin_admin.dart init-keys --print-env` and `init-custodian`; store the printed secrets in a password manager.
3. Push the repository and apply `render.yaml` as a Render Blueprint, pasting `DATABASE_URL`, `ECHOPIN_KEYS`, `SUPABASE_URL`, `SUPABASE_SERVICE_KEY`, `BREVO_API_KEY`, `MAIL_FROM` and `CUSTODIAN_PUBLIC_KEYS`. The server validates every setting at start-up and names the one that is wrong.
4. Create the first admin: `dart run bin/echopin_admin.dart create-admin --email you@example.org --name "Your Name"` with `DATABASE_URL` and `ECHOPIN_KEYS` set; sign in on the app and turn on two-step.
5. Set the GitHub repository variable `ECHOPIN_SERVER_URL` so the keep-awake workflow pings the free instance every 10 minutes.

Self-hosted: `cd server && docker compose up` (PostgreSQL included), or `dart run bin/server.dart` with a `.env` from `.env.example`. Details, configuration reference and the CLI are in `docs/SERVER.md`.

5.5.6 First run checklist

1. `GET https://your-server/v1/health` answers `{"ok":true,"database":"up"}`.
2. Sign up in the app; the 8-digit code arrives by email.
3. Choose an account PIN; open **Account**.
4. Admin: turn on two-step, open the **Console**, check **Backups** the next day and press **Check chain** under **Audit**.
5. Responders: import the custodian key under Console → key icon.

5.5.7 Uninstall and data removal

- Phone: uninstalling removes the device key, PINs and history (the keystore entries go with the app). The account itself is removed by an admin in Console → People.
- Server: data retention is automatic (records 365 days, diagnostics 90, closed reports 730); the audit log is never deleted. Accounts that have opened records cannot be deleted by design (their openings must remain attributable); disable them instead.

Appendix: verification commands

```
 flutter analyze
dart format --output=none --set-exit-if-changed lib test integration_test
flutter test
(cd packages/echopin_protocol && dart test)
(cd server && tool/test.sh)
(cd server && tool/live_check.sh)
xvfb-run -a flutter test integration_test/lan_transport_test.dart -d linux
flutter build apk --release && flutter build linux --release && flutter build web --release
```